

# Two Simulations for Mystery Tokens

Unit 3 · Topic 3.14 · TODAY'S PROBLEM

Suppose a company claims 10,000 mystery tokens are 50% red, 30% teal, and 20% gold. An SRS of 80 has the shown counts.

**Leo:** “Mix and redraw these same 80 tokens without replacement; recompute  $\chi^2$  for 1,000 trials.”

**Amara:** “For each trial, generate 80 independent colors with probabilities 0.50, 0.30, and 0.20, then recompute  $\chi^2$ .” In Amara’s trials, 186 statistics are at least the observed value.

Token sample

| Color | Claim | Observed |
|-------|-------|----------|
| Red   | 0.50  | 32       |
| Teal  | 0.30  | 30       |
| Gold  | 0.20  | 18       |

## FIRST TAKE (5 min, on your sheet)

Which simulation can reveal whether the sample’s color discrepancies are unexpected? State what varies in one trial.

- DOK 1** (a) Calculate the three expected counts under the company’s claimed color model.
- DOK 2** (b) Compute the  $\chi^2$  contributions and total, give  $df$ , estimate the upper tail from Amara’s trials, and check 10 percent and expected counts.
- DOK 3** (c) Choose the simulation that answers the chance-variation question and judge surprise at 5 percent. Compare what each trial fixes or varies, use the simulated tail, explain the rejected output’s false assurance, and avoid claiming the model is proved.

Video + follow-along: [u8\\_lesson1\\_live.html](https://u8_lesson1_live.html) (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

